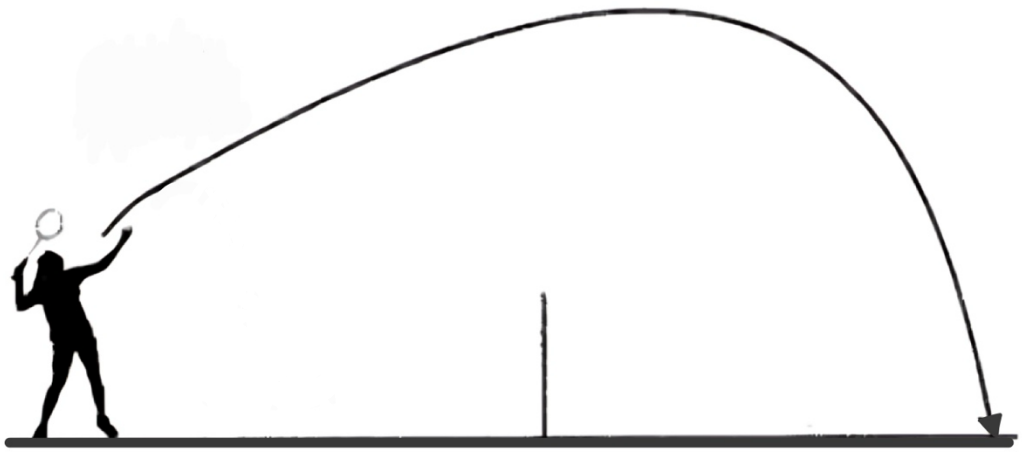
Modelling and Optimization

How can mathematical models be used to determine the optimal angles between the badminton shuttlecock and racket for a clear shot in badminton?



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Introduction

Badminton is a sport played with a racket and shuttlecock, against one or two people. I've played badminton for many years now with an external coach, focusing on mastering various shots, and each of them play a crucial role in the game, enhancing my ability to hit anywhere. She showed me how I am supposed to hit the shuttle for it to move in a certain way by changing the angle of my racket contacting the shuttlecock.

A clear shot is used when you are at the edge on your side of the court, to send the shuttlecock high and landing at the opposite side into the opponent's court, for getting out of defensive positions, giving you time to reposition by forcing the opponent to move to the back to the court and give up space at the front.

However, there is one challenge I often face during matches. I tend to lose points when my clear shots are not deep enough in the enemy's court, giving them an opportunity to smash the shuttle back at me.

I want to overcome this obstacle by understanding and practicing the optimal angle of a clear shot more consistently. By improving my accuracy, I believe I can minimize mistakes and become a more effective player.

Aim of Exploration

Within this investigation, I aim to model the trajectory of a clear shot using formulas for projectile motion, finding the derivative and graphing the derived equations to determine the optimal angle to reach the intended position on the other side of the court. To do this, I will integrate to derive the velocity and displacement functions, and create a horizontal range function to find the optimal angle to maximize distance.

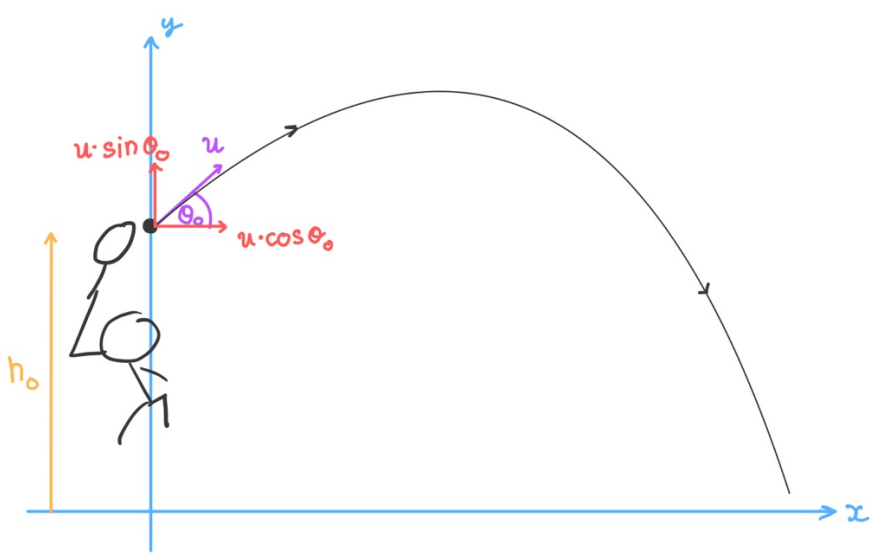
Assumptions

The main assumption of the model is that I will only be considering two dimensions: length (x) and height (y) directions. This means that there will not be any cross-court shots, lowering the distance travelled. This allows the model and calculations to be simpler.

Preliminary Understanding

For the model, the player will hit the shuttlecock h_0 metres from the ground, hit with an initial speed of u, with angle of elevation of θ_0. Because the velocity is at an angle of θ_0 to horizontal, the vectors will be split up into horizontal (x) and vertical (y), **where the positive is to the right and upwards respectively**. The black dot is where the shuttle was hit initially.

Figure 1: Shuttlecock Trajectory with initial velocity (without forces)



However, applying projectile motion will also come with drag force, which is a force that acts opposite to the direction of the velocity it is travelling, and the formula is referred below:

$$D = \frac{1}{2}\rho C_d A v^2$$

Table 1: Drag Force Formula

Variable	**Description**	**Unit**
D	Drag force	Newton (N)
ρ	Density of air	kg/m^3
C_d	Coefficient of drag	No units
A	Cross-sectional area	m^2
v	Velocity	m/s

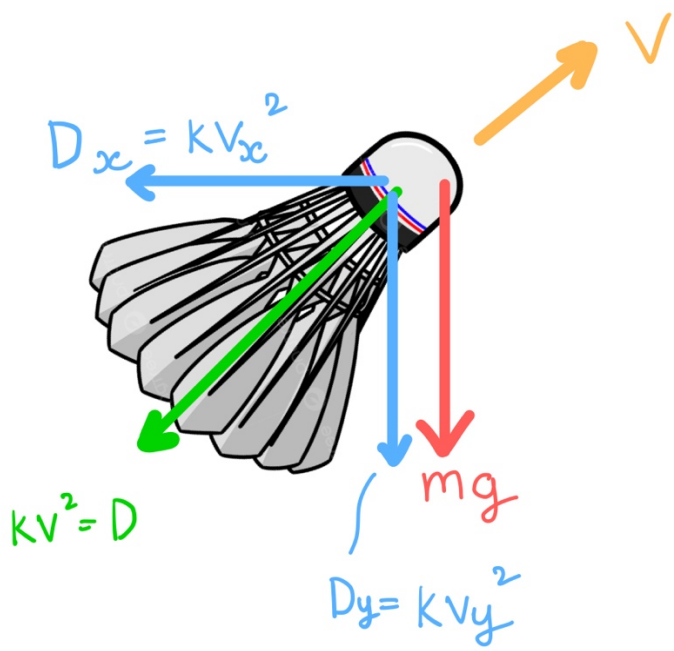
However, looking at the drag force formula above, ρ, C_d, and A are constant with time, so let:

$$k = \rho C_d A \text{ where k is a positive constant}$$

$$\therefore D = \frac{1}{2}kv^2$$

When deriving the forces of motions, it is important to draw a free body diagram, which is "used to show the relative magnitude and direction of all forces acting upon an object in a given situation" (The Physics Classroom). Similar to vector calculations, since the shuttlecock is hit at an angle, the forces must be divided into x and y axes. **The free body diagram below shows forces and velocities of a shuttlecock at an arbitrary point during its flight.**

Figure 2: Free Body Diagram of Shuttlecock



D_x and D_y shows the drag force component in the x and y axes respectively, represented in blue lines from the original drag force shown in green. There is always gravity pointing downwards on an object, and is calculated by the mass times gravitational field strength. This force is usually where the center of mass is, but since the cork is where most of the weight is, it will be placed there.

It can be observed that the direction of shuttlecock is changing during its flight through air. This indicates that velocity is changing and not constant, meaning the shuttlecock is accelerating. Hence, Newton's second law of motion is required to express the equation of motion of accelerated object.

$$F = ma$$

Table 2: Newton's Second Law

Variable	**Description**	**Unit**
F	Net / Resultant force	N
m	Mass of object	kg
a	Acceleration	m/s^2

Finding Horizontal Velocity: $v_x(t)$

By looking at the force acting along x-axis in the free body diagram, there is only one force acting on the object which is F_x or $-k{v_x}^2$. Hence, there must also be an acceleration acting along x-axis which is a_x or $\frac{dv_x}{dt}$.

$$F_x = D_x = ma_x$$

$$-k{v_x}^2 = m \cdot \frac{dv_x}{dt}$$

Integrating by doing separable differential equations,

$$\int -\frac{k}{m} \cdot dt = \int \frac{1}{{v_x}^2} \cdot dv_x$$

$$-\frac{k}{m}t + C_1 = -\frac{1}{v_x}$$

$$\frac{k}{m}t + C_2 = \frac{1}{v_x} \text{ where } C_2 = -C_1$$

$$v_x = \frac{1}{\frac{k}{m}t + C_2}$$

Substituting initial condition: at $t = 0$, $v_x = u \cdot cos\theta_0$

$$u \cdot cos\theta_0 = \frac{1}{0 + C_2}$$

$$v_x(t) = \frac{1}{\frac{k}{m}t + \frac{1}{u \cdot cos\theta_0}} \qquad \text{[EQ.1]}$$

Finding Vertical Velocity: $v_y(t)$

For the y direction, F_y and a_y will be the net force and acceleration of the free body diagram within the y axis.

$$F_y = D_y - mg = ma_y$$

$$-k{v_y}^2 - mg = m \cdot \frac{dv_y}{dt}$$

$$-k\left({v_y}^2 + \frac{mg}{k}\right) = m \cdot \frac{dv_y}{dt}$$

$$\int -\frac{k}{m} \cdot dt = \int \frac{1}{{v_y}^2 + \frac{mg}{k}} \cdot dv_y$$

$$-\frac{k}{m}t + C = \frac{1}{\sqrt{\frac{mg}{k}}} \cdot tan^{-1}\left(\frac{v_y}{\sqrt{\frac{mg}{k}}}\right)$$

$$\tan\left(-\sqrt{\frac{kg}{m}} \cdot t + C\right) = \frac{v_y}{\sqrt{\frac{mg}{k}}}$$

$$v_y = \sqrt{\frac{mg}{k}} \cdot \tan\left(-\sqrt{\frac{kg}{m}}t + C\right)$$

Substituting initial conditions: at $t = 0$, $v_y = u \cdot \sin(\theta_0)$

$$u \cdot \sin(\theta_0) = \sqrt{\frac{mg}{k}} \cdot \tan(0 + C)$$

$$C = arctan\left(\sqrt{\frac{k}{mg}} \cdot u \cdot sin\theta_0\right)$$

$$v_y(t) = \sqrt{\frac{mg}{k}} \cdot \tan\left(-\sqrt{\frac{kg}{m}}t + arctan\left(\sqrt{\frac{k}{mg}} \cdot u \cdot sin\theta_0\right)\right) \quad \text{[EQ.2]}$$

Finding Horizontal Displacement: x(t)

To solve for function of x(t) and y(t), $v_x(t)$ from [EQ.1] can be written as $\frac{dx}{dt}$ and $v_y(t)$ can be written as $\frac{dy}{dt}$.

$$v_x(t) = \frac{1}{\frac{k}{m}t + \frac{1}{u \cdot cos\theta_0}} = \frac{dx}{dt}$$

$$\int \frac{dt}{\frac{k}{m}t + \frac{1}{u \cdot cos\theta_0}} = \int dx$$

$$x = \ln\left|\frac{k}{m}t + \frac{1}{u \cdot cos\theta_0}\right| \cdot \frac{m}{k} + C$$

Substitute initial conditions at $t = 0$, $x = 0$

$$0 = \frac{m}{k}\ln\left|\frac{1}{u \cdot cos\theta_0}\right| + C$$

$$C = -\frac{m}{k}\ln\left|\frac{1}{u \cdot cos\theta_0}\right|$$

$$x(t) = \frac{m}{k} \cdot \ln\left|\frac{k}{m}t + \frac{1}{u \cdot cos\theta_0}\right| - \frac{m}{k}\ln\left|\frac{1}{u \cdot cos\theta_0}\right|$$

u is speed, meaning it is only the magnitude of the initial velocity, hence it's scalar, so $0 < u$. Since we hit within the angles $0 \leq \theta_0 \leq 90$, $0 \leq \cos\theta_0$, therefore $0 < k,\ 0 < m$ and $0 < t$.

$$\therefore x(t) = \frac{m}{k} \cdot \ln\left(\frac{k}{m}t + \frac{1}{u \cdot cos\theta_0}\right) - \frac{m}{k}\ln\left(\frac{1}{u \cdot cos\theta_0}\right)$$

$$x(t) = \frac{m}{k} \cdot \ln\left(\frac{k}{m}t \cdot u \cdot cos\theta_0 + 1\right) \quad \text{[EQ.3]}$$

Finding Vertical Displacement: y(t)

In the y direction, it is the same method, continuing from [EQ.2]:

$$v_y(t) = \sqrt{\frac{mg}{k}} \cdot \tan\left[-\sqrt{\frac{kg}{m}}\,t + arctan\left(\sqrt{\frac{k}{mg}} \cdot u \cdot sin\theta_0\right)\right] = \frac{dy}{dt}$$

$$\int \sqrt{\frac{mg}{k}} \tan\left[-\sqrt{\frac{kg}{m}} \cdot t \ + \arctan\left(\sqrt{\frac{k}{mg}} \cdot u \cdot sin\theta_0\right)\right] \cdot dt = \int dy$$

<u>Simplify into these variables:</u>

$$\boldsymbol{\alpha} = -\sqrt{\frac{kg}{m}}$$

$$\boldsymbol{\beta} = \arctan\left(\sqrt{\frac{k}{mg}} \cdot u \cdot sin\theta_0\right)$$

$$\boldsymbol{\gamma} = \sqrt{\frac{mg}{k}}$$

$$\therefore y(t) = \gamma \int \tan(\alpha t + \beta)\, dt = \gamma \cdot \left(-\frac{\ln\left(\cos(\alpha t + \beta)\right)}{\alpha}\right) + C$$

Substitute initial conditions at t=0, y=h_0

$$h_0 = -\gamma \cdot \frac{\ln(\cos\beta)}{\alpha} + C$$

$$C = h_0 + \gamma \cdot \frac{\ln(cos\beta)}{a}$$

$$\therefore y(t) = h_0 + \frac{\gamma}{\alpha} \ln\left|\frac{\cos\,(\beta)}{\cos\,(\alpha t + \beta)}\right| \qquad \text{[EQ.4]}$$

Graphing Shuttlecock Trajectory

After obtaining the formula, I put in the constant into [EQ.3] and [EQ.4] using Table 3 such as the mass of shuttlecock and the initial speed.

Table 3: Constants Table

Constants	Symbol	Value
Mass of shuttlecock	m	0.005 kg
Gravitational field strength	g	9.81 ms^{-2}
Initial speed	u	60.0 ms^{-1}
Initial angle	θ	0.349 rad
Initial height of contact	h_0	2.90 m
Constant of drags (calculated in appendix)	k	0.00123
Alpha (seen in page 8)	α	-1.555
Beta (seen in page 8)	β	1.272
Gamma (seen in page 8)	γ	6.308

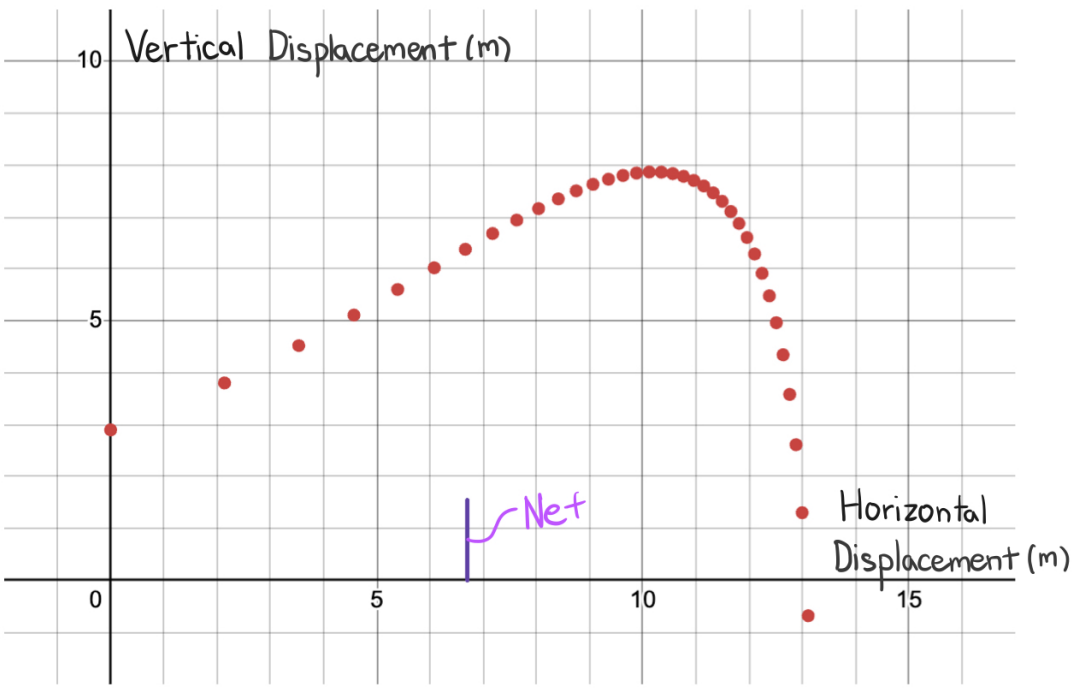
The constants except k, α, β and γ are found from the averages of the player statistics. The average indicates the typically performance of the players, including factors such as player height, racket height, and initial speed. These are essential for accurately modeling the motion of the shuttlecock in play. From these constants, I am able to find the x and y axis of the trajectory of the shuttlecock as it travels across to the other side.

Table 4: Sample data of the trajectory of shuttlecock with given constants above

(Rest in Appendix)

Time (s)	x-position: x(t) (m) By substitute into [EQ.3]	y-position: y(t) (m) By substitute into [EQ.4]
0.00	0	2.900
0.05	2.140	3.804
0.10	3.534	4.522
0.15	4.570	5.111
…	…	…
0.80	10.12	7.866
…	…	…
1.75	13.10	-0.6794

Graph 1: Trajectory of shuttlecock

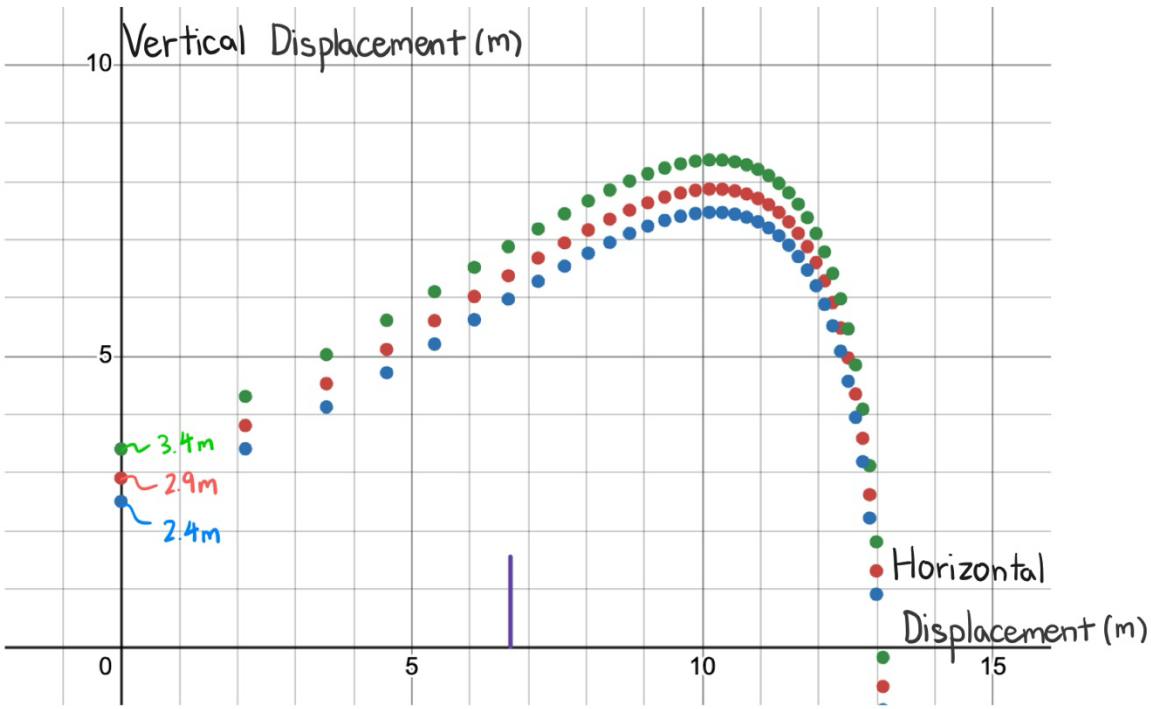


From Graph 1 above, when x is in between 0 to 10 m, the graph has positive slope or increasing function. In words, the shuttlecock can travel 10 m from where the player hits it. When x is in between 10 to 13 m, the graph has negative slope or decreasing function. It means that when x increases, y decreases. Therefore, the shuttlecock lands at x=13 within 1.75 seconds. That means the opponent would need to run to the back of the court within 1.75 seconds to the back of the court to hit back. The purple line indicates the net, and it clears it over with ease.

Extension: Change in Initial Height (h_0)

My coach always tells me to hit the shuttlecock from as high as possible, but from my experience, the contact point did not change the outcome. This section is to see if my coach is right or not. I will investigate this by changing the h_0 to 2.4m, 2.9m, and 3.4m, in 0.5m intervals, I investigated the extent of how the height of the players would influence the trajectory.

Graph 2: Height Difference Trajectory



I observed three main aspects of the trajectories:

1. The peak height would increase as the h_0 increases.
2. The general shape stays the same.
3. The X displacement when it reaches the ground (Y displacement=0) is nearly identical

The general conclusion is that the initial height does not hinder the horizontal displacement travelled. However, the vertical displacement did increase. Hence, my coach would be correct. This exploration could also apply to player heights, as badminton have varying heights from 1.65m to 2.00m. However, naturally the taller players would have longer arms, resulting in longer reaches, and comparing clear shots alone does not have a significant impact in comparing height differences.

Finding Optimal Angle for Horizontal Range: R(θ_0)

Range (R) is defined as the horizontal distance that the shuttlecock travels at the point when it hits the ground. The optimal range for a clear shot is the shuttlecock landing at the furthest back of the badminton court, also while not going out, which is 13.41m.

To find the horizontal distance (range), first determine the time that the shuttlecock hits the floor, hence when $y(t) = 0$.

$$y(t) = h_0 + \frac{\gamma}{\alpha} \ln \left| \frac{\cos(\beta)}{\cos(\alpha t + \beta)} \right| = 0$$

$$\ln \left| \frac{\cos(\beta)}{\cos(\alpha t + \beta)} \right| = -\frac{a}{\gamma} h_0$$

$$\frac{\cos(\beta)}{\cos(\alpha t + \beta)} = e^{-\frac{a h_0}{\gamma}}$$

$$e^{\frac{a h_0}{\gamma}} \cos(\beta) = \cos(\alpha t + \beta)$$

$$\alpha t + \beta = \arccos\left(e^{\frac{a h_0}{\gamma}} \cos(\beta)\right)$$

$$t = \frac{\arccos\left(e^{\frac{a h_0}{\gamma}} \cos(\beta)\right) - \beta}{\alpha}$$

Here, t is the time that the shuttlecock hits the floor. This also depends on θ_0, which is part of β.

To find the horizontal location at the point when hitting the floor, substitute into [EQ.3]

$$x(t) = \frac{m}{k} \cdot \ln\left(\frac{k}{m} t \cdot u \cdot cos\theta_0 + 1\right)$$

This function with the change in θ_0 will be called R(θ_0):

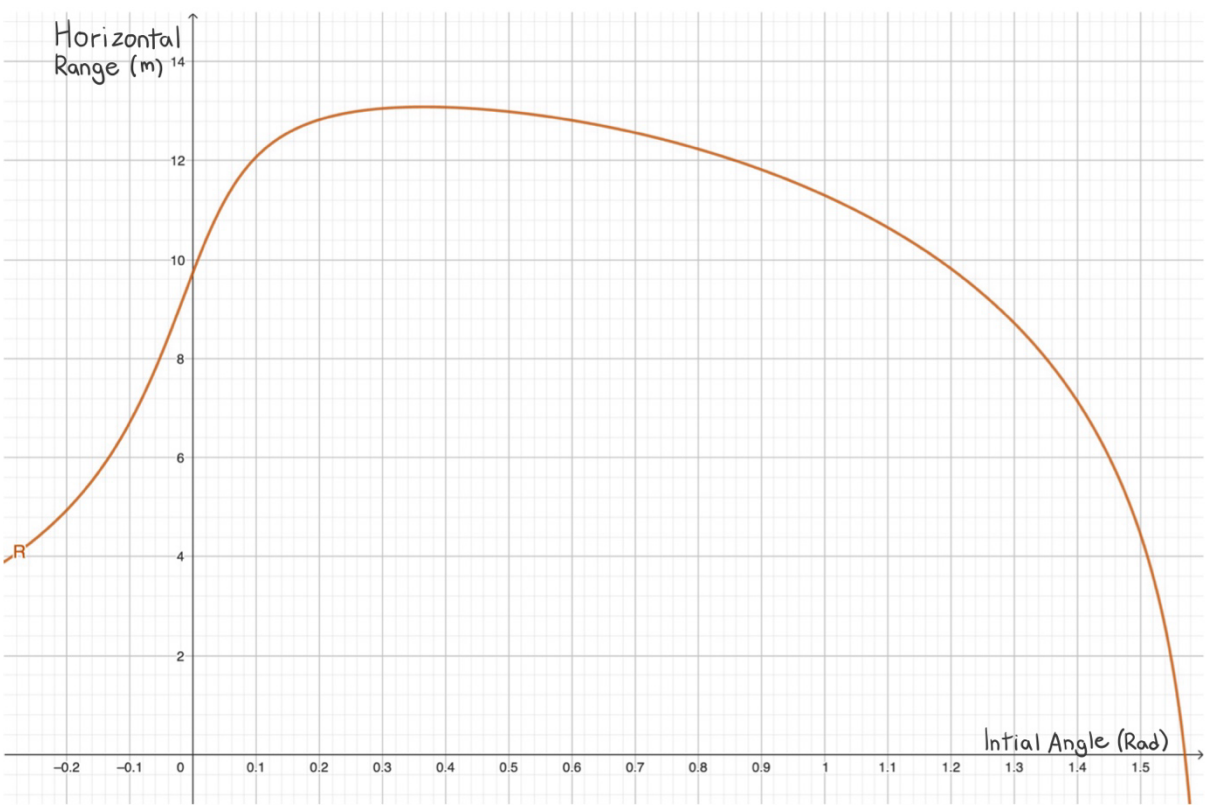
$$Range \equiv R(\theta_0) = \frac{m}{k} \times \ln\left(\frac{k}{m} \times u \times cos\theta_0 \times \frac{\arccos\left(e^{\frac{a h_0}{\gamma}} \cos(\beta)\right) - \beta}{\alpha} + 1\right)$$

Table 5: Sample Data of Relationship between Initial Angle and Horizontal Range (m)

(Rest in Appendix)

Initial Angle (deg)	Initial Angle (rad)	Horizontal Range (m)
0	0.000	9.755
1	0.01745	10.31
2	0.03491	10.82
3	0.05236	11.25
4	0.06981	11.61
5	0.08727	11.90
6	0.10472	12.13
7	0.12217	12.32
8	0.13963	12.48
…	…	…

Graph 3: Horizontal Range R(θ_0) vs Initial Angle (rad)



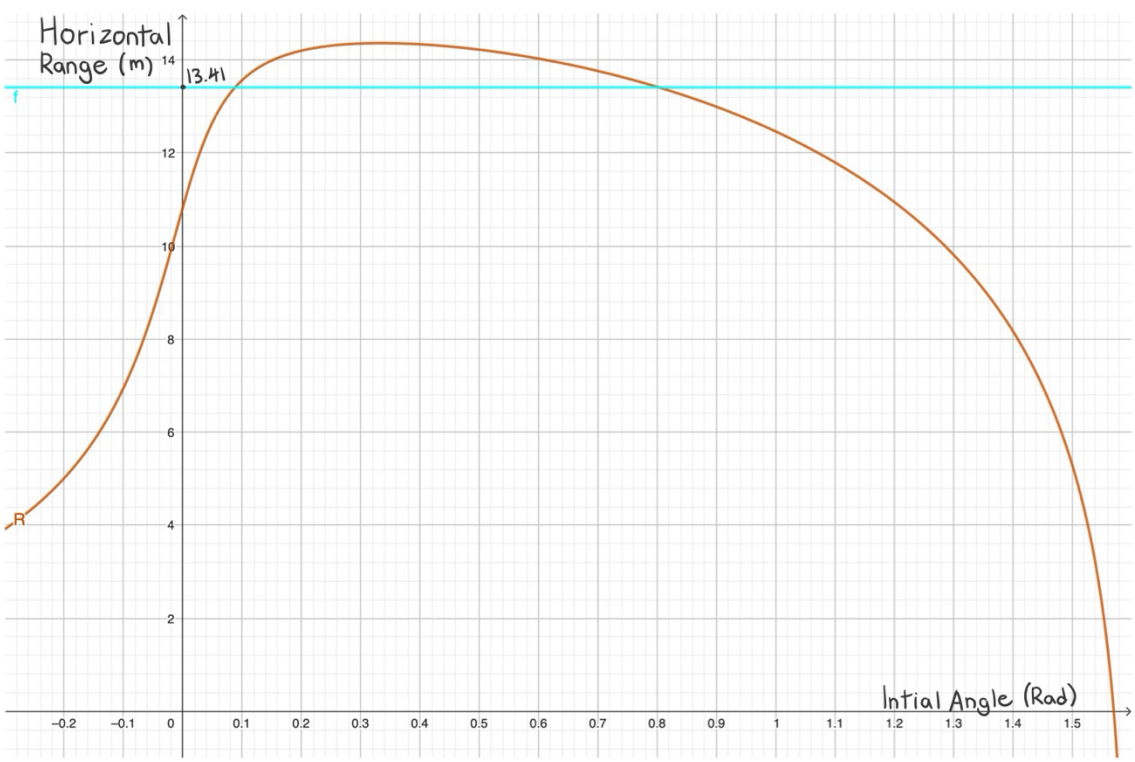
To verify the function, $R(90) = 0$, meaning if you hit the shuttlecock with initial angle of 90°, the range is zero. This gives us confidence that the equation derived make sense.
By finding the maximum point on the graph using GeoGebra where $0 \leq \theta_0 \leq \pi$,
$\boldsymbol{\theta_0 = 0.3659\, rad \approx 20.9°}$.

When substituting 20.9 ° into R (θ_0), $\mathrm{R}(20.9) = 13.08$, which is lower than the length of the court, which is 13.41. This shows that within the set of constants above in Table 3, the optimal angle is 20.9°.

Extension: Changing Initial Velocity

The initial velocity of the badminton shuttle could vary depending on person to person, depending on positioning, height, or strength.
For example, when the initial velocity (u) is 80m/s, it will look like the following graph. The neon blue line represents the maximum distance that the shuttlecock can travel (13.41m) without going out of the court.

Graph 4: Horizontal Range R (θ_0) vs Initial Angle (rad) when u = 80



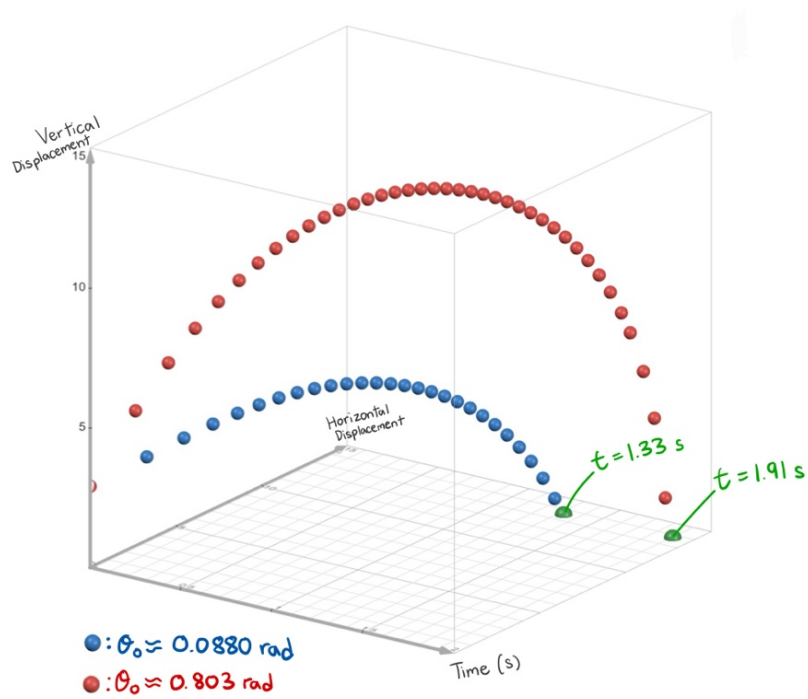
The local maximum point on this graph is (0.334, 14.4), meaning if the person hits the shuttlecock at 80m/s at an angle of 0.334 rad, it will horizontally go 14.4m, and goes out of the court. So, by finding out where $R(\theta_0)$ and $y = 13.41$ intersects, the horizontal range can be found.

$$R(\theta_0) = 13.41$$

$$\theta_0 \approx 0.0880 \text{ rad}, 0.803 \text{ rad}$$

Using the horizontal and vertical displacement equations [EQ. 3] and [EQ. 4] respectively, we can find the better shape of a clear shot.

Graph 5: Comparison between initial angles (Blue: $\theta_0 \approx 0.0880$ rad, Red: $\theta_0 \approx 0.803$ rad)



To re-iterate, a clear shot is supposed to go far **horizontally and vertically**. The blue trajectory does not go as high vertically, and in a real game, it allows the opponent to hit the shuttlecock back earlier. This is also evident by when both shuttle reaches the ground, where in the blue, it hits the floor in 1.33 seconds, while the red hits the floor in 1.91 seconds seen in the green orbs. In this comparison, $\theta_0 \approx 0.803$ rad would be the **optimal initial angle**.

Evaluation

Evaluation of Method

For the method, I was able to incorporate the effects of drag force into the trajectory function of the model. This enhances the usability of the model by accounting for the resistance the shuttlecock experiences as it moves through the air.

However, for the shuttlecock property, I assumed that the shuttlecock will always travel with the cork facing the motion of direction, meaning that the drag coefficient is constant throughout the whole motion while the shuttlecock is in the air. In reality, the orientation of the shuttlecock could change midair, influencing the drag characteristics and lead to unpredictable changes in trajectory. This results in the velocity, and consequently, the displacement functions to have a higher value compared to reality, resulting in the shuttlecock peaking at 8m in height, which is usually not possible.

Evaluation of Application

I modeled a clear shot when a player would be jumping up and hitting the shuttle. However, a clear shot would include more motion, as the opponent would have to hit it back at you at the very back, and when hitting the shuttle back, the player must stretch their shoulders and snap it back to generate the force. This creates human error as the force would decrease throughout the match. In reality, replicating these specific shots with the correct velocity with the

Conclusion

In the end, I was able to get the optimal angle desired to hit the shuttlecock to the intended position on the other side of the court. By solving simultaneous differential equations, I could accurately model the motion of the shuttlecock by integrating from the acceleration in both the x and y directions to the corresponding displacements. This mathematical model provides a useful framework for understanding the dynamics of a badminton shot and highlights the importance of precise angle and force control in competitive play.

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Appendix

Table 3: Constants Table

Density of air	ρ	1.293 kgm^3
Coefficient of drag	C_d	0.525
Cross-sectional area of shuttle	A	0.00363 m^2

Table 4: The trajectory of shuttlecock

Time	x-position	y-position
0.00	0	2.90000
0.05	2.140414459	3.80409
0.10	3.534196804	4.52247
0.15	4.569600756	5.11141
0.20	5.393730323	5.60394
0.25	6.078364388	6.02102
0.30	6.663960067	6.37680
0.35	7.175582917	6.68120
0.40	7.629842873	6.94141
0.45	8.038315905	7.16279
0.50	8.409392842	7.34937
0.55	8.749350351	7.50422
0.60	9.063006484	7.62967
0.65	9.354140311	7.72750
0.70	9.625770729	7.79902
0.75	9.880347591	7.84513
0.80	10.11988631	7.86641
0.85	10.34606486	7.86312
0.90	10.56029515	7.83523
0.95	10.76377648	7.78238
1.00	10.95753615	7.70392
1.05	11.1424609	7.59883
1.10	11.31932141	7.46570

1.15	11.48879172	7.30261
1.20	11.65146475	7.10704
1.25	11.8078648	6.87570
1.30	11.95845776	6.60424
1.35	12.10365946	6.28687
1.40	12.24384261	5.91578
1.45	12.37934257	5.48010
1.50	12.51046215	4.96417
1.55	12.63747573	4.34448
1.60	12.76063266	3.58327
1.65	12.88016028	2.61442
1.70	12.99626638	1.30533
1.75	13.10914139	-0.67935
1.80	13.21896031	-4.79188

Table 5: Relationship between Initial Angle and Range

Initial angle (deg)	Initial angle (rad)	Beta	Range
0	0.000	0.00000	9.755466698
1	0.017	0.16450	10.31377338
2	0.035	0.32050	10.8158687
3	0.052	0.46188	11.24691509
4	0.070	0.58580	11.60507275
5	0.087	0.69217	11.89710657
6	0.105	0.78250	12.13327734
7	0.122	0.85898	12.32400439
8	0.140	0.92383	12.47837323
9	0.157	0.97906	12.60375872
10	0.175	1.02638	12.70594588
11	0.192	1.06719	12.78940709
12	0.209	1.10262	12.85758722
13	0.227	1.13359	12.91314591
14	0.244	1.16083	12.95814826

15	0.262	1.18495	12.99420952
16	0.279	1.20640	13.02260332
17	0.297	1.22560	13.04434221
18	0.314	1.24285	13.06023762
19	0.332	1.25844	13.07094468
20	0.349	1.27257	13.07699601
21	0.367	1.28543	13.07882728
22	0.384	1.29718	13.07679679
23	0.401	1.30795	13.07120051
24	0.419	1.31785	13.06228379
25	0.436	1.32697	13.05025043
26	0.454	1.33541	13.03526996
27	0.471	1.34323	13.01748328
28	0.489	1.35049	12.99700722
29	0.506	1.35724	12.97393823
30	0.524	1.36354	12.94835525
31	0.541	1.36943	12.92032213
32	0.559	1.37494	12.88988956
33	0.576	1.38010	12.85709663
34	0.593	1.38495	12.82197207
35	0.611	1.38951	12.78453536
36	0.628	1.39380	12.74479748
37	0.646	1.39784	12.70276166
38	0.663	1.40166	12.65842382
39	0.681	1.40526	12.61177306
40	0.698	1.40867	12.56279192
41	0.716	1.41189	12.51145659
42	0.733	1.41495	12.45773706
43	0.750	1.41784	12.40159716
44	0.768	1.42059	12.34299454
45	0.785	1.42319	12.2818806
46	0.803	1.42567	12.21820032

47	0.820	1.42802	12.15189207
48	0.838	1.43025	12.08288728
49	0.855	1.43238	12.01111013
50	0.873	1.43440	11.93647707
51	0.890	1.43633	11.85889637
52	0.908	1.43816	11.77826745
53	0.925	1.43990	11.6944802
54	0.942	1.44157	11.60741419
55	0.960	1.44315	11.51693766
56	0.977	1.44465	11.42290653
57	0.995	1.44609	11.32516301
58	1.012	1.44745	11.22353428
59	1.030	1.44875	11.11783069
60	1.047	1.44999	11.00784388
61	1.065	1.45116	10.89334445
62	1.082	1.45228	10.77407935
63	1.100	1.45334	10.64976872
64	1.117	1.45435	10.52010222
65	1.134	1.45531	10.38473471
66	1.152	1.45621	10.24328101
67	1.169	1.45707	10.09530978
68	1.187	1.45789	9.940335976
69	1.204	1.45865	9.77781182
70	1.222	1.45938	9.607115702
71	1.239	1.46006	9.427538511
72	1.257	1.46070	9.23826662
73	1.274	1.46130	9.038360507
74	1.292	1.46186	8.826727596
75	1.309	1.46238	8.602087319
76	1.326	1.46286	8.362925599
77	1.344	1.46331	8.107434618
78	1.361	1.46372	7.833431811

79	1.379	1.46410	7.538248817
80	1.396	1.46444	7.218575951
81	1.414	1.46475	6.870238951
82	1.431	1.46502	6.487869192
83	1.449	1.46526	6.064399889
84	1.466	1.46547	5.590264942
85	1.484	1.46565	5.052061318
86	1.501	1.46579	4.430176229
87	1.518	1.46590	3.694238613
88	1.536	1.46598	2.793451495
89	1.553	1.46603	1.632823811
90	1.571	1.46605	0